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6.5.R1 (1/1 point)

You are doing a simulation in order to compare the effect of using Cross-Validation or a Validation set. For each iteration of the simulation, you generate new data and then use both Cross-Validation and a Validation set in order to determine the optimal number of predictors. Which of the following is most likely?

- ☐ The Cross-Validation method will result in a higher variance of optimal number of predictors
- ☒ The Validation set method will result in a higher variance of optimal number of predictors ✓
- ☐ Both methods will produce results with the same variance of optimal number of predictors
- ☐ Not enough information is given to decide

EXPLANATION

Cross-Validation is similar to doing a Validation set multiple times and then averaging the answers. As such, we expect it to have lower variance than the Validation set method. This is why Cross-Validation is appealing (especially for small n).

[Hide Answer](#)*You have used 1 of 5 submissions*

